

Project Title
Intelligent Job Portal

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT :

Modern job portals contain thousands of job postings, making manual browsing slow and inefficient for applicants. This project presents an Intelligent Job Portal that uses efficient data structures and algorithms to provide fast, skill- and location-aware job search. Job postings are indexed using hashing and trie-based structures for efficient skill and title lookup, while set intersection is used to combine skill, location, and other filtering criteria.

A max-heap is employed to rank matching jobs according to a weighted relevance score, allowing the most suitable listings to be displayed first. The system also supports fast autocomplete and scalable searching through indexed data. The proposed approach aims to reduce irrelevant results, improve search speed and relevance, and maintain efficient performance as the number of job postings increases. By combining these techniques, the portal provides applicants with a streamlined search experience and helps them identify relevant opportunities more effectively.

Signature of the Guide